

Tutorial No : 4

Tutorial Title: Write a program in C++ to perform arithmetic operations on complex numbers. Design the class for complex number representation and the operations to be performed. The objective of this assignment is to learn the concepts of classes and objects.

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Batch:C

Program:

```
#include <iostream>
using namespace std;
```

```
class Complex
```

```
{
    int r, i;
```

```
public:
```

```
    void input()
```

```
{
    cout << "Enter real part: ";
    cin >> r;
```

```
    cout << "Enter imaginary part: ";
    cin >> i;
```

```
}
```

```
    void show()
```

```
{
    if (i >= 0)
        cout << r << " + " << i << "i" << "\n";
    else
        cout << r << " - " << -i << "i" << "\n";
}
```

```
Complex add(Complex c)
```

```
{
    Complex ans;
    ans.r = r + c.r;
```

```
    ans.i = i + c.i;  
    return ans;  
}
```

```
Complex subtract(Complex c)  
{  
    Complex ans;  
    ans.r = r - c.r;  
    ans.i = i - c.i;  
    return ans;  
}
```

```
Complex multiply(Complex c)  
{  
    Complex ans;  
    ans.r = (r * c.r) - (i * c.i);  
    ans.i = (r * c.i) + (i * c.r);  
    return ans;  
}
```

```
Complex divide(Complex c)  
{  
    Complex ans;  
    int den = (c.r * c.r) + (c.i * c.i);  
  
    ans.r = ((r * c.r) + (i * c.i)) / den;  
    ans.i = ((i * c.r) - (r * c.i)) / den;  
  
    return ans;  
}  
};
```

```
int main()  
{  
    Complex c1, c2, ans;  
  
    cout << "Enter first complex number:\n";
```

```
c1.input();

cout << "\nEnter second complex number:\n";
c2.input();

ans = c1.add(c2);
cout << "\nAddition = ";
ans.show();

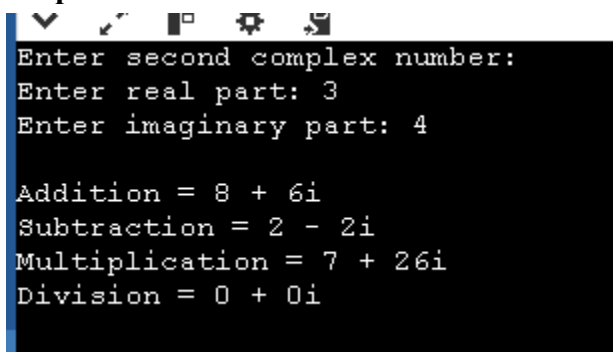
ans = c1.subtract(c2);
cout << "Subtraction = ";
ans.show();

ans = c1.multiply(c2);
cout << "Multiplication = ";
ans.show();

ans = c1.divide(c2);
cout << "Division = ";
ans.show();

return 0;
}
```

Output:



The screenshot shows a terminal window with a black background and white text. At the top, there is a toolbar with icons for a dropdown menu, a cursor, a square, a gear, and a magnifying glass. The terminal output is as follows:

```
Enter second complex number:
Enter real part: 3
Enter imaginary part: 4

Addition = 8 + 6i
Subtraction = 2 - 2i
Multiplication = 7 + 26i
Division = 0 + 0i
```